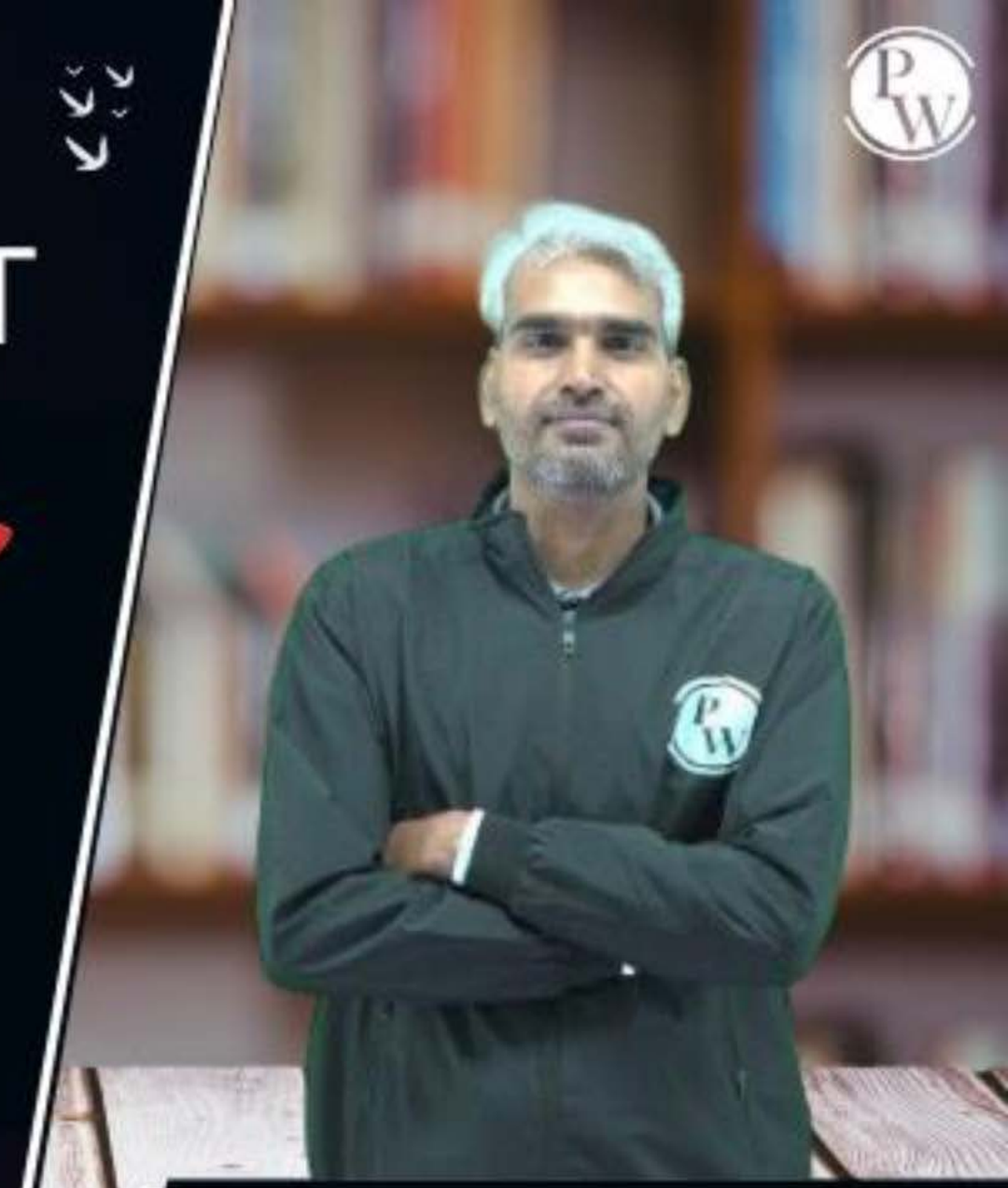
Computer Science & IT



Discrete Mathematics

Set Theory & Algebra

Lecture No. 08



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Recap of Previous Lecture



- Topic — Reflexive relation
- Topic — Irreflexive relation
- Topic — Symmetric relation
- Topic — Anti-symmetric relation
- Topic — Asymmetric relation
- Topic — Transitive relation

$|A| = n$

2^{n^2-n}

2^{n^2-n}

$2^n \cdot 2^{\frac{n^2-n}{2}} = 2^{\frac{n(n+1)}{2}}$

$2^n \cdot 3^{\frac{n^2-n}{2}}$

$3^{\frac{n^2-n}{2}}$

Slide

Topics to be Covered



- Topic — Transitive relation
- Topic — Reflexive closure
- Topic — Symmetric closure
- Topic — Transitive closure

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Topic : Transitive Relation

- A relation R on set A is called transitive relation if and only if

if $(a \overset{R}{=} b$ and $b \overset{R}{<} c)$ then $a \overset{R}{<} c$, $\forall a, b, c \in A$

i.e. if $((a,b) \in R$ and $(b,c) \in R)$ then $(a,c) \in R$, $\forall a, b, c \in A$

Slide

eg:- Let $A = \{1, 2, 3\}$

① $R_1 = \{\}$ Empty relation is the smallest transitive rel^n on any set A.

② $R_2 = A \times A$ 'AxA' is the largest transitive rel^n on any set A.

③ $R_3 = \{(1,1), (2,2)\}$ It is a transitive rel^n

Because of presence of diagonal order pairs, there will be no problem w.r.t. transitivity.

Note Let $(a,a) \in R$

$\therefore$ $(a,a) \in R$ and $(a,a) \in R$, then because of transitivity (a,a) must be present in rel^n R;

and it is already a member of R

$\therefore$ No problem

Note: Let $(a,a) \in R$ & $(a,b) \in R$

i.e. $(a,a) \in R$ and $(a,b) \in R$, then

because of transitivity (a,b) must be present in the rel^n, and it is already a member of R.

$\therefore$ No problem

Note: Let $(b,a) \in R$ & $(a,a) \in R$

i.e. $(b,a) \in R$ and $(a,a) \in R$, then

because of transitivity (b,a) must be present in rel^n R, and (b,a) is already a member of R,

$\therefore$ No problem.

H.W

let $A = \{1, 2, 3\}$

④ $R_4 = \{(1,2), (2,1), (1,1)\}$ Check whether R_4 is transitive or not

(1,1) is diagonal order pair, ∴ No problem

it Ends with '2' ⇒ ∴ We only need to check w.r.t. order pairs that starts with '2'. {and (2,1) is only such order pair}

$(1,2) \in R_4$ and $(2,1) \in R_4$ ∴ By transitivity (1,1) must belong to R_4
and $(1,1) \in R_4$ ∴ No problem

it Ends with '1' ∴ We need to check with order pairs that starts with '1', (1,2) and (1,1) are such order pairs

- (2,1) and (1,1) ∴ By transitivity (2,1) must belong to rel^n and it is already present, ∴ No problem
- (2,1) and (1,2) ∴ By transitivity (2,2) must be present in rel^n but $(2,2) \notin R_4$ ∴ R_4 is not transitive

H.W

eg. $A = \{1, 2, 3\}$

$R_5 = \{(1,2), (1,3)\}$ transitive or not ?

it Ends with '2' $\Rightarrow$ ∴ we need to check with order pairs that starts with '2', and no order pair starts with '2' ∴ Hence it will not create any problem w.r.t transitivity

it Ends with '3' ∴ we need to check with order pairs that starts with '3', and no order pair starts with '3' ∴ Hence it will not create any problem w.r.t transitivity

No order pair dis-satisfy the Condⁿ of transitivity

∴ R_5 is a transitive Relⁿ

H.W.

eg: $A = \{1, 2, 3\}$

$R_6 = \{(2,3), (1,2)\}$ transitive or not ?

Ends with '3' → No order pair starts with '3' ∴ No Problem

Ends with '2' ∴ check with order pairs that starts with 2, (i.e, (2,3))

$(1,2) \in R_6$ & $(2,3) \in R_6$ ∴ By transitivity

(1,3) must belong to R_6

But $(1,3) \notin R_6$

∴ R_6 is not transitive

H.W

eg 1: Let $A = \{1, 2, 3\}$

$R = \{(1,1), (1,2), (1,3), (2,2), (3,1)\}$ transitive or not

Ends with '1'

∴ Check with ordered Pairs that Starts with '1'

$(3,1)$ and
- $(1,1)$ ∴ $(3,1)$ already in Ref
- $(1,2)$ ∴ $(3,2)$, but $(3,2) \notin R$ ∴ Not transitive
- $(1,3)$ ∴ $(3,3)$, but $(3,3) \notin R$ ∴ Not transitive

H.W.

eg 2: Let $A = \{1, 2, 3\}$

$R = \{(1,1), (1,3), (2,2), (3,1), (3,3)\}$ transitive or not ?

it is a transitive rel^n

eg. Let $A = \{1, 2, 3\}$

$R = \{(1,1), (1,3), (2,2), (3,1), (3,2)\}$ transitive or not?

$(1,3) \in R$ & $(3,2) \in R$ $\therefore$ $(1,2)$

But $(1,2) \notin R$

$\therefore$ Not transitive

eg. Let $A = \{a, b, c, d\}$

$R = \{(a,d), (b,a), (b,c), (c,a), (c,d), (d,c)\}$ transitive or not?



$(a,d) \in R$ and $(d,c) \in R$ $\therefore$ (a,c)

But $(a,c) \notin R$

$\therefore$ Not transitive



Topic : Reflexive Closure

- Let R is any relation on set A.
- Reflexive closure of relation R is the smallest reflexive relation on set A that contains relation R

eg. let $A = \{1, 2, 3\}$

$R = \{(1,1), (1,2), (2,1), (3,2), (3,3)\}$

Reflexive Closure of R = $\{(1,1), (1,2), (2,1), (3,2), (3,3), (2,2)\}$

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Note :- ① If given relation R is already a reflexive relation on set A, then reflexive closure of relation R will be relation R itself.

② Reflexive closure of $R = R \cup \Delta_A$

Where R is any relation on set A



Topic : Symmetric Closure

- Let R is any relation on set A,
- Symmetric closure of relation R is the smallest symmetric relation on set A that contains R

eg: Let $A = \{1, 2, 3\}$

$R = \{(1,1), (1,2), (2,1), (2,2), (3,1), (3,2)\}$

Symmetric Closure of R $= \{(1,1), (1,2), (2,1), (2,2), (3,1), (3,2), (1,3), (2,3)\}$

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Note: { ① If R is symmetric relation on set A, then symmetric closure of relation R will be relation R itself



Topic : Transitive Closure

- Let R is any relation on set A, Transitive closure of relation R is the smallest transitive relation on set A that contains relation R

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Note:

If relation R on set A is already a transitive relation, then transitive Closure of relation R will be relation R itself.

eg 1: Let $A = \{1, 2, 3\}$

$R = \{(1,1), (1,2), (1,3), (2,2), (3,1)\}$

Find transitive closure of R

eg 1: Let $A = \{1, 2, 3\}$

$R = \{(1,1), (1,2), (1,3), (2,2), (3,1)\}$

Given, $R = \{(1,1), (1,2), (1,3), (2,2), (3,1)\}$

1st iteration → $R^* = \{(1,1), (1,2), (1,3), (2,2), (3,1) \mid (3,2) (3,3)\}$ ✓

2nd iteration → $R^{**} = \{(1,1), (1,2), (1,3), (2,2), (3,1), (3,2), (3,3)\}$

In this iteration, no new order pair needs to be added to satisfy transitivity

We will keep iterating, until we reach to a situation when no new order pair needs to be added in order to satisfy transitivity

And the relation w.r.t. last iteration (i.e. the iteration in which no new order pair is added) will be the transitive Closure of given relation

eg 2: Let $A = \{1, 2, 3\}$

$R = \{(1,1), (1,3), (2,2), (3,1), (3,3)\}$

Find transitive closure of R

$R = \{(1,1), (1,3), (2,2), (3,1), (3,3)\}$

1st iteration

$R^* = \{(1,1), (1,3), (2,2), (3,1), (3,3)\}$ ...

No new order pair needs to be added

Transitive closure of R is R^*

if $\boxed{R^* = R}$ i.e, given relation is a transitive relation

eg. Let $A = \{1, 2, 3\}$

$R = \{(1,1), (1,3), (2,2), (3,1), (3,2)\}$

Find transitive closure of R.

$R = \{(1,1)\ (1,3), (2,2), (3,1), (3,2)\}$

1st iteration → $R^* = \{(1,1)\ (1,3), (2,2), (3,1), (3,2) \mid (1,2)\ (3,3)\}$

Rearranged R^* (it is not iteration) $= \{(1,1)\ (1,2)\ (1,3), (2,2), (3,1), (3,2)\ (3,3)\}$

$R^{**} = \{(1,1)\ (1,2)\ (1,3), (2,2), (3,1), (3,2)\ (3,3) \mid$

No new order pair needs to be added

∴ R^{**} is transitive Closure of R

H.W.

e.g. Let $A = \{a, b, c, d\}$

$R = \{(a,d), (b,a), (b,c), (c,a), (c,d), (d,c)\}$

Find transitive closure of R



2 mins Summary

Topic Transitive relation

Topic Reflexive closure

Topic Symmetric closure

Topic Transitive closure

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THANK - YOU